ADAPT: Adaptive Dual-projection Architecture for Perceptive Traversal

***Abstract*— Agile humanoid locomotion in complex 3D environments requires reasoning about both ground-support geometry and body-level traversable space under the strict latency constraints of high-frequency control. Existing perceptive locomotion methods often face two coupled limitations: compact 2.5D maps are efficient but miss vertical clearance constraints, whereas dense 3D representations provide richer geometry at higher computational cost; meanwhile, fixed sensing scale cannot adapt to changes in motion speed and local environmental complexity. We propose ADAPT (Adaptive Dual-projection Architecture for Perceptive Traversal), a perception-control framework that addresses both representation efficiency and sensing-scale adaptation. ADAPT decomposes local 3D geometry into two compact projections: a horizontal elevation map for terrain support and a vertical distance map for body-level clearance. It further treats the sensing radius of the vertical map as a reinforcement learning policy action, enabling the robot to expand its perceptual horizon for long-range anticipation and contract it for higher local resolution in cluttered spaces. Experiments show that ADAPT achieves a 94.7% success rate across diverse 3D traversal tasks, reduces observation dimensionality by 98.6% compared with voxel-based alternatives, and reduces computational overhead through adaptive sensing. The learned policy achieves zero-shot transfer to a Unitree G1 humanoid robot and enables real-world space traversal.**

I. Introduction

Humanoid robots are expected to move through human-scale environments that contain stairs, cluttered passages, protruding structures and other three-dimensional constraints. For successful traversal, the controller depends on not only estimating where the feet can step on, but also understanding whether the whole body can safely pass through the surrounding space, including overhead and lateral obstacles that may cause collision or loss of balance. This makes perceptive traversal a coupled problem of terrain understanding, body-level clearance reasoning and real-time whole-body control.

However, onboard perception for legged robots is constrained by strict latency and bandwidth requirements. Existing representations therefore tend to emphasize either efficiency or geometric completeness. Conventional 2.5D elevation maps [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11] provide compact terrain geometry for foothold reasoning, but they are intrinsically blind to overhangs, side walls and body-level clearance constraints. Depth-based representations [12], [13], [14], [15], [16], [17], [18], [19], [20] offer direct visual observations, yet they can be affected by view changes, occlusion and illumination variation. Volumetric approaches [21], [22], [23] encode richer 3D geometry, but their dimensionality and processing cost make high-frequency policy learning and deployment difficult. These limitations necessitate a representation that preserves 3D traversability while remaining computationally compact [24], [25], [26].

A second limitation is that perception is usually acquired with a fixed spatial range [21], [9], [4], which is misaligned with the changing information demand of locomotion: fast motion needs a longer horizon while cluttered traversal needs higher local resolution. A perceptive controller should therefore decide not only how to act given perception, but also how much spatial context should be sensed.

To address these two issues, this paper proposes **ADAPT** (**A**daptive **D**ual-projection **A**rchitecture for **P**erceptive **T**raversal) that restructures local 3D geometry into two orthogonal and information-dense projections. A horizontal elevation map describes ground geometry for stable foothold selection, while a vertical distance map describes body-level traversable space for avoiding overhangs and obstacles. Instead of using a fixed perceptual horizon, ADAPT further exposes the sensing radius of the vertical distance map to the reinforcement learning policy as a control variable. The policy can thus expand its perceptual horizon during fast traversal and contract it for fine-grained local reasoning in cluttered scenes. In this way, ADAPT jointly addresses what spatial information should be represented and how much of it should be sensed at each moment.

Empirical evaluation demonstrates that this design improves both robustness and efficiency. ADAPT achieves a 94.7% success rate across a series of 3D negotiation tasks and surpasses baselines with fixed sensing ranges by 30%. The dual-projection architecture reduces observation dimensionality by 98.6% and decreases the training time per iteration to 32.7% of that required by voxel-based alternatives. The learned adaptive sensing mechanism further reduces computational overhead by 56% by matching resource consumption to motion complexity. These results indicate that ADAPT is not only a compact perception representation, but also a perception-control coupling mechanism suitable for real-time humanoid traversal.

The main contributions of this work are:

- Proposing a highly efficient formulation that decouples complex 3D environments into compact projections. This paradigm reduces observational dimensionality while rigorously preserving essential geometric constraints, thereby ensuring the high-frequency perception strictly required for agile locomotion.
- Introducing a bidirectional closed-loop mechanism that embeds perceptual configurations directly into the reinforcement learning action space. This formulation empowers the robot to autonomously modulate its sensory settings based on instantaneous dynamic states and

environmental complexity, thereby achieving an optimal trade-off between navigational safety and precise spatial perception.
- Demonstrating robust zero-shot sim-to-real transfer on a Unitree G1 humanoid robot with successful navigation through diverse structural challenges.

II. Related Works

A. Perception for Legged Locomotion

Perceptive legged locomotion has increasingly relied on environment representations that allow a policy or planner to anticipate terrain changes before contact. The central design question is how to encode enough geometry for safe locomotion while keeping the observation efficient. As summarized in Table I, existing methods can be grouped into elevation-map, image/depth-based, and volumetric or point-based representations. These categories differ in what spatial structure they preserve and what computational burden they impose on the controller.

TABLE I: ADAPT comparison with existing works

Method	Perceptual Representation	Native Sampling	Sensory Dimension	Overhead	Adaptive Sensing
PIM [4]	Elevation Map	0.10 m	96	✗	✗
Beamdojo [5]	Elevation Map	0.10 m	225	✗	✗
Vb-com [6]	Elevation Map	0.10 m	225	✗	✗
Parkour [14]	Depth Image	48×64 px	3072	✓	✗
Omni [23]	Point Cloud	0.05π rad	251	✓	✗
Gallant [21]	Voxel Grid	0.05 m	40960	✓	✗
HumanoidPF [27]	Joint Position	39 dims	39	✓	✗
ADAPT (ours)	Dual Projection	**0.025π rad**	**578**	✓	✓

Elevation maps [1], [2], [4], [5], [6], [7], [11] provide efficient 2.5D grids for foothold reasoning. However, their height-field assumption inherently discards vertical structures like overhangs and side walls, missing the body-level clearance constraints crucial for humanoid traversal in 3D clutter.

Depth and vision-based methods [13], [12], [14], [15], [16], [18], [19], [20] offer richer scene observations for agile locomotion. Yet, their unstructured inputs heavily burden the learning system and are susceptible to occlusions, viewpoint shifts, and illumination changes. While reduced visual encodings mitigate this, they tend to entangle ground geometry with visual noise rather than explicitly separating the needed spatial constraints.

Volumetric, point-cloud and multi-plane representations aim to recover more complete 3D spatial awareness [9], [21], [22], [23], [27]. These methods are better suited to collision avoidance and constrained-space traversal because they can encode obstacles beyond the terrain surface. Their main drawback is the cost of maintaining and processing dense 3D structure. Voxel grids scale cubically with resolution and incur high memory traffic [24], leading to severe inference latency (e.g., $4.4\,\mathrm{Hz}$ in [25]) that fails the $\geq 10\,\mathrm{Hz}$ requirement for closed-loop control [26] and inflated training iteration times (e.g., $> 22\,\mathrm{s/iter}$ in [21]). Point-based representations require careful sampling and aggregation to remain stable for policy learning. Multi-plane methods mitigate the cubic cost but leave vertical clearance implicit, forcing the policy to infer the gap between an overhead obstacle and the supporting ground across plane indices rather than from a direct distance measurement. The increased latency and observation dimensionality can slow training and reduce the responsiveness needed for high-speed humanoid motion.

ADAPT is motivated by the gap between efficient but incomplete 2.5D maps and complete but expensive 3D representations. Instead of representing the whole local volume densely, it decomposes traversability into two task-oriented projections: a horizontal elevation map for ground support and a vertical distance map for body-level clearance. This design preserves the spatial constraints most relevant to humanoid traversal while avoiding the redundancy of dense voxel observations.

B. Adaptive Perception and Behavior Modulation

The need for adaptive perception arises from the non-uniform information demand of locomotion: fast traversal requires a long perceptual horizon to anticipate obstacles, whereas cluttered traversal benefits more from high-resolution local geometry. A fixed sensing range cannot satisfy both simultaneously.

Spatial attention and foveation methods select informative regions or suppress irrelevant visual content [28], [29], while multi-range mapping assigns different densities to nearby and distant regions [30]. However, these mechanisms adapt perception after the sensing structure has been specified; the spatial extent and resolution levels remain fixed by design rather than continuously selected by the policy.

Active perception mechanisms change the sensing focus or viewpoint according to task context [31], and perceptive locomotion methods adapt gait and footholds based on terrain observations [14], [4], [5], [6]. Nevertheless, perception is typically organized as an upstream module with a fixed range, and the downstream policy adapts actions to those observations. The missing component is a feedback path in which the same policy that generates motion also regulates the amount of spatial information it receives.

ADAPT fills this gap by making the sensing radius an explicit policy action and feeding the previous radius back as part of the observation. Unlike active-viewpoint methods, ADAPT does not require changing the physical sensor pose; instead, it learns a policy-controlled sensing range for the constructed vertical map. This formulation turns perceptual horizon selection from a predefined sensing schedule into a closed-loop control decision. Through end-to-end reinforcement learning, the robot learns to expand the vertical-map range for long-horizon anticipation and to contract it when fine local resolution is more valuable. ADAPT therefore couples behavior modulation with information-intake modulation, enabling the perceptual scale to change with motion speed and environmental complexity during 3D humanoid traversal.

III. METHODS

A. Problem Formulation

The perceptive traversal task is formalized as a goal-conditioned Partially Observable Markov Decision Process (POMDP). At each time step t, the policy π_θ receives a partial observation $\mathbf{o}_t$ encapsulating the robot's kinematic state and local environmental context. It then outputs a composite action $\mathbf{a}_t = (\mathbf{q}_t, r_t)$ that simultaneously dictates the joint position targets $\mathbf{q}_t$ and an adaptive perception radius r_t. This dual-action space seamlessly couples physical actuation with dynamic sensory intake.

Driven by the stochastic transition dynamics $\mathcal{T}$, the agent seeks to learn an optimal policy π^* that maximizes the expected cumulative discounted return:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t \mathcal{R}(s_t, a_t) \right], \tag{1}$$

where $\gamma \in (0, 1)$ is the discount factor and T is the horizon. The scalar reward $\mathcal{R}(s_t, a_t)$ encodes multiple objectives, including velocity tracking, energy efficiency, and safety constraints.

B. Training Pipeline

The training pipeline is illustrated in Fig. 1.

1) Observation Space: The observation vector o_t integrates proprioceptive feedback, task commands and exteroceptive data, which is defined as

$$\mathbf{o}_t = \{\mathbf{c}_t, \mathbf{v}_t, \boldsymbol{\omega}_t, \mathbf{g}_t, \mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}, \mathbf{E}_t, \mathbf{D}_t, r_{t-1}\}. \tag{2}$$

The specific components in (2) are:

- *Task Command:* The velocity command $\mathbf{c}_t \in \mathbb{R}^3$.
- *Proprioceptive States:* These states include base linear velocity $\mathbf{v}_t$, base angular velocity $\boldsymbol{\omega}_t$, the gravitational acceleration $\mathbf{g}_t$, joint positions $\mathbf{q}_t$, joint velocities $\dot{\mathbf{q}}_t$ and the preceding action $\mathbf{a}_{t-1}$.
- *Exteroceptive States:* The actor receives synthetic point cloud data during learning, acquired via the LiDAR simulation framework from Wang et al. [23]. We rasterize the raw point cloud into grid structures to facilitate efficient processing. Consequently, this input is processed into two orthogonal robot-centric maps as shown in Fig. 1: (1) A *horizontal elevation map* ($\mathbf{E}_t \in \mathbb{R}^{21\times17}$) that captures terrain geometry via discrete height sampling. (2) A *vertical distance map* ($\mathbf{D}_t \in \mathbb{R}^{13\times17}$) that encodes spatial traversability. The distance map is generated by projecting points into a cylindrical sector spanning from -45 to $+45$ degrees around the vertical axis, where each point records the distance to the nearest obstacle.
- *Adaptive Perception Radius:* The radius r_{t-1} acts as a dynamic gain modulating the field of view for both projection maps. It is constrained to $r_t \in [1, 5]$ m throughout training and deployment. The vertical distance map uses a cylindrical sector grid with a fixed number of bins $(N_r, N_\theta) = (13, 17)$ as illustrated in Fig. 2. For a selected radius r_{t-1}, the radial sampling interval is defined as $\delta_r = r_{t-1}/N_r$, which makes the grid size grow linearly with the sensing radius. Therefore, a larger radius extends the perceptual horizon but coarsens sampling resolution, while a smaller radius concentrates the grids and yields finer resolution. Consequently, modulating the sensing radius reduces computation overhead and minimizes distal sensor noise.

The critic operates on a privileged observation space to facilitate accurate estimation of the value function V_t. In addition to the actor's input vector o_t, the critic receives the noise-free ground-truth elevation map $\hat{\mathbf{E}}_t$ directly acquired from the simulation engine. All components are then scaled, flattened and concatenated into a 648-dimensional vector to serve as input to the recurrent policy.

2) Action Space: The policy produces a composite action $\mu_t \in \mathbb{R}^{13}$ for each step at 50 Hz to simultaneously handle motor control and perceptual modulation. The composite action a_t is thus formally defined as

$$a_t = \{q_t, r_t\} \tag{3}$$

The specific components in (3) are:

- *Joint Targets:* The first 12 elements of μ_t specify target joint positions. They are scaled and superimposed onto a nominal configuration q_{default} to yield the action targets $q_t \in \mathbb{R}^{12}$ defined as

$$q_t = s_a \cdot \mu_{t,1:12} + q_{\text{default}} \tag{4}$$

  where s_a is the action scale. These targets are then executed by a low-level Proportional-Derivative (PD) controller.
- *Adaptive Perception Radius:* The final element of μ_t dictates the perception radius $r_t \in [1, 5]$ m for the subsequent timestep, which is directly used to shape the exteroceptive map based on current locomotion dynamics. This range is empirically determined: a radius below 1 m provides insufficient lookahead for timely obstacle avoidance maneuvers, whereas a radius exceeding 5 m introduces excessive distal noise that degrades traversability estimation. Both extremes severely impair collision avoidance performance.

3) Reward Functions: The reward function $\mathcal{R}$ integrates multiple objectives to encourage robust and efficient locomotion adhering to safety constraints. The total reward $r_t = \sum_i w_i r_i(t)$ is a weighted sum of the following components:

- *Motion Control:* Precise velocity tracking, postural stability, and energy efficiency are promoted by penalizing deviations in base orientation, excessive joint velocities or accelerations, and high-frequency action oscillations.
- *Safety Mechanisms:* Safe and prolonged operation is ensured by penalizing critical failures such as body collisions, mesh interpenetrations, and joint limit violations. Furthermore, survival bonuses and foot airtime rewards are provided to foster dynamic and robust obstacle clearance.
- *Active Perception:* The active sensing behavior is optimized by dynamically adapting the perception radius

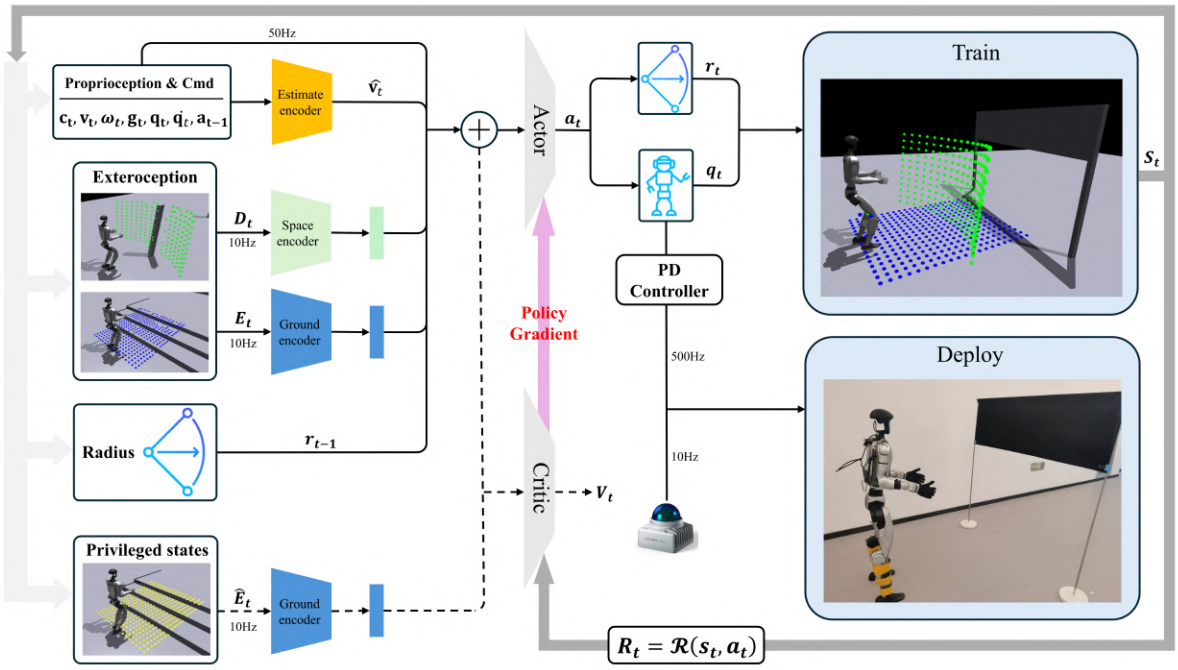


Fig. 1: Illustration of the training architecture for the proposed control framework ADAPT.

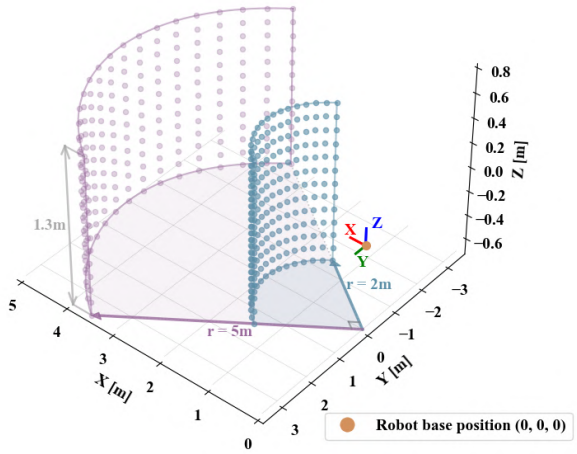


Fig. 2: Illustration of the adaptive perception radius mechanism. The vertical distance map is constructed on a cylindrical sector with a fixed number of cells (17×13). A smaller radius (blue) provides higher spatial resolution for precise near-field sensing, while a larger radius (purple) extends the perception range at the cost of reduced resolution. The policy learns to dynamically adjust this radius based on the robot's locomotion state.

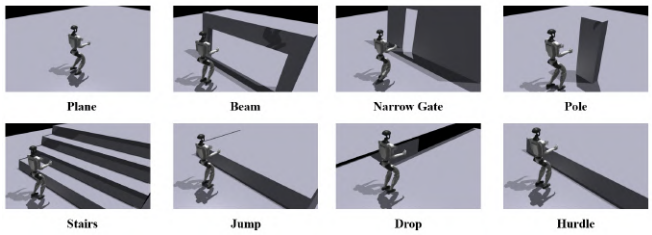


Fig. 3: Examples of procedurally generated obstacles in our *BarrierTrack* training environment. The curriculum includes a diverse set of challenges such as stairs, drop, hurdles, and constrained passages like beams and narrow gates, promoting the development of a versatile locomotion policy.

based on the robot's velocity and proximity to obstacles. Additional penalties on abrupt variations and boundary saturations ensure smooth and stable sensor adjustments.

4) Terrain Design: BarrierTrack (Fig. 3), a customized environment assembling discrete obstacle modules, is designed to cultivate versatile locomotion. Each episode features a sequence of five obstacles of the same terrain type with progressively increasing difficulty according to a curriculum schedule.

A curriculum-based strategy is adopted where each terrain type τ is parameterized by a difficulty scalar $s \in [0, 1]$, and its generation parameters are interpolated as $\mathbf{p}_\tau(s) = (1 - s)\,\mathbf{p}_\tau^{\min} + s\,\mathbf{p}_\tau^{\max}$ based on Table II. Perlin noise is superimposed onto ground surfaces to simulate terrain irregularity and prevent overfitting.

C. Training Architecture

As shown in Fig. 1, the training framework employs an asymmetric recurrent actor-critic architecture with a Gated Recurrent Unit (GRU) to capture short-horizon temporal dependencies during locomotion. This framework allows the critic to leverage privileged simulation states to accelerate convergence. The critic combines the recurrent states with noise-free ground truth including elevation to enable precise

TABLE II: Parameters for generating curriculum training terrains. The ↑ symbol indicates the parameter value increases as difficulty scales up ($\mathbf{p}_\tau^{\min} < \mathbf{p}_\tau^{\max}$), while the ↓ symbol indicates the value decreases as the task becomes harder ($\mathbf{p}_\tau^{\min} > \mathbf{p}_\tau^{\max}$).

Terrain	Param. (m)	$\mathbf{p}_\tau^{\min}$	$\mathbf{p}_\tau^{\max}$
Beam	Clearance Height ↓	1.33	1.10
	Beam Depth ↑	0.20	0.20
Narrow Gate	Gate Width ↓	0.80	0.45
	Position Offset ↑	0.00	0.60
Pole	Pole Width ↓	0.50	0.10
	Position Offset ↑	0.00	0.00
Stairs	Step Height ↑	0.10	0.30
	Step Length ↓	0.50	0.30
Jump	Wall Height ↑	0.05	0.40
	Wall Depth ↑	0.10	0.30
Drop	Drop Height ↑	0.10	0.60
	Platform Length ↓	0.50	0.30
Hurdle	Hurdle Height ↑	0.05	0.40
	Hurdle Depth ↑	0.05	0.10

value estimation. In contrast, the actor operates exclusively on deployable observations. The policy is optimized using the Proximal Policy Optimization (PPO) algorithm. Note that the observation vector o_t is processed through two separate MLP encoders, which independently encode the exteroceptive inputs $\mathbf{E}_t$ and $\mathbf{D}_t$ into 32-dimensional latent variables. These latent representations are concatenated to form a 64-dimensional environmental embedding and fused with the proprioceptive states before being passed to the actor and critic heads.

A distinguishing feature of the proposed architecture is the closed-loop modulation of perceptual parameters. Conditioned on the recurrent hidden state, the actor outputs the perception radius r_t to tightly couple the sensing range with robot states and obstacle proximity.

IV. Experiments

A. Experimental Configuration

1) Simulation Environment: All policies are trained using NVIDIA Isaac Gym [32] on an NVIDIA RTX 4090 GPU with a 12-DoF Unitree G1 humanoid robot. LiDAR point clouds are synthesized using the simulator from [23], which can accurately model sensor noises and occlusion from static terrains and dynamic robot links.

2) Real-world Environment: The trained policy is then deployed on a physical Unitree G1 robot. A Livox Mid-360 LiDAR is mounted on the head to provide raw point cloud data. State estimation and point cloud registration are handled by Fast-LIO2 [33]. Both the horizontal and vertical elevation maps $\mathbf{E}_t$ are extracted from the registered point cloud using the GPU-accelerated pipeline from [3] at 10 Hz. The map generation follows the exact same procedure used in simulation. Joint commands are executed at 50 Hz on an onboard NVIDIA Jetson Orin NX.

B. Simulation Experiments

1) Evaluation Setup: All methods are evaluated on the highest difficulty tier of *BarrierTrack*. The benchmark includes all seven terrain primitives, grouped into two categories: terrains that require precise foot placement like Stairs, Drop, Jump and Hurdle, and the others that require traversable-space selection like Beam, Pole and Narrow Gate. For each terrain type, every episode contains five obstacles of the same class with progressively increasing difficulty; obstacle parameters are sampled within the range given in Table II according to the curriculum schedule.

2) Metrics: Robot performance is evaluated by the mean and standard deviations over two metrics: (1) Success Rate (R_{succ}), which is the fraction of discrete obstacles that the robot successfully navigates within a single trial, i.e., the number of obstacles crossed divided by 5. and (2) Normalized Traversal Progress (R_{trav}), which is the forward distance traveled along the course normalized by the total track length. Note that R_{trav} is typically no lower than R_{succ}, as the robot may stumble forward for some distance even after failing to clear an obstacle, thereby accumulating partial traversal progress.

3) Performance Comparison: ADAPT is benchmarked against four baselines to evaluate its navigation capability and computational efficiency. Each method is evaluated across all seven terrain types under three speed commands: $v \in \{0.5, 1.0, 1.5\}$ m/s. For each terrain-speed combination, 10 trials are conducted and the average results are reported. The four selected baselines are:

- PIM [4]: A single-stage training method that uses an elevation map from LiDAR point cloud as exteroceptive input. Contrastive learning aligns the generated elevation map with ground truth.
- Parkour [14]: A two-stage method where a teacher policy is first trained with ground-truth elevation maps and then agile locomotion skills are transferred through supervised distillation to a student policy that only uses depth images.
- TSDF: A camera-based approach that performs 3D reconstruction via nvblox [34] and feeds local grid TSDF values as exteroceptive input for policy training.
- Gallant [21]: A voxel-based approach that provides full 3D awareness but at a higher computational cost.

Table III details the performance comparison, while Table IV presents the associated training costs. ADAPT (Ours) achieves the highest success rates across all terrains while requiring the lowest training overhead. The performance of the baselines is consistent with the sensing characteristics within each method. PIM relies on a 2.5D elevation map and shows reduced performance on Pole and Narrow Gate terrains where vertical and lateral clearance cues are important. Parkour uses depth images and performs less reliably on terrains requiring broader spatial coverage, such as Jump, Hurdle and Beam. TSDF provides volumetric awareness, but its performance remains limited on fine-grained terrains although incurring the highest training cost. Gallant improves

TABLE III: Benchmarked Comparison in Simulation. R_{succ}: success rate (%, ↑). R_{trav}: normalized traversal progress (%, ↑).

Method	Plane		Stairs		Drop		Jump	
	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}
PIM [4]	100 (±0.0)	100 (±0.0)	65.33 (±37.57)	70.00 (±34.61)	99.33 (±3.59)	99.77 (±1.26)	90.67 (±16.92)	91.67 (±14.62)
Parkour [14]	100 (±0.0)	100 (±0.0)	28.00 (±28.10)	37.20 (±24.93)	91.33 (±14.31)	92.40 (±13.52)	44.67 (±20.45)	51.40 (±17.49)
TSDF	100 (±0.0)	100 (±0.0)	40.00 (±20.00)	47.43 (±17.50)	56.67 (±24.81)	63.33 (±22.75)	43.33 (±13.74)	48.63 (±13.65)
Gallant [21]	100 (±0.0)	100 (±0.0)	70.00 (±22.95)	76.30 (±21.75)	80.00 (±31.41)	83.23 (±27.62)	68.00 (±13.27)	74.10 (±11.34)
Ours	**100** (±0.0)	**100** (±0.0)	**94.67** (±13.60)	**95.90** (±10.63)	**100.00** (±0.00)	**100.00** (±0.00)	**98.67** (±4.99)	**98.90** (±4.12)
Method	Hurdle		Beam		Pole		Narrow Gate	
	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}
PIM [4]	82.00 (±20.88)	86.37 (±16.44)	66.67 (±9.43)	75.53 (±5.28)	2.67 (±6.80)	6.17 (±7.09)	27.33 (±15.04)	31.17 (±14.17)
Parkour [14]	47.33 (±13.15)	52.67 (±12.81)	14.00 (±11.72)	20.53 (±6.93)	5.33 (±8.84)	15.00 (±7.90)	48.00 (±16.81)	53.03 (±16.05)
TSDF	49.33 (±11.23)	55.47 (±13.98)	33.33 (±11.93)	40.83 (±11.37)	7.33 (±9.64)	14.37 (±9.07)	22.67 (±16.11)	29.80 (±14.31)
Gallant [21]	54.67 (±16.28)	62.47 (±14.21)	32.00 (±25.61)	40.40 (±27.17)	46.00 (±44.17)	49.40 (±41.82)	66.00 (±25.90)	69.57 (±23.35)
Ours	**90.00** (±14.38)	**91.80** (±12.80)	**96.67** (±7.45)	**97.57** (±5.47)	**98.00** (±6.00)	**98.57** (±4.79)	**80.00** (±18.62)	**83.03** (±16.30)

TABLE IV: Training Efficiency Comparison. Train/iter: wall-clock time per training iteration (s, ↓). GPU peak: peak GPU memory usage (MB, ↓). ↑/↓ indicates that higher/lower values are better.

Method	Train/iter (s, ↓)	GPU Peak (MB, ↓)
PIM [4]	2.6	10485
Parkour [14]	97.8	28791
Gallant [21]	34.6	36054
TSDF	246	38473
Ours	**11.3**	**15870**

spatial coverage relative to the other baselines, but its higher-dimensional voxel input requires higher computational cost and achieves lower success rates than ADAPT on most terrains. Overall, ADAPT provides a favorable balance between traversal performance and computation efficiency across various tasks.

As shown in Table IV, PIM achieves the lowest training cost among all methods owing to its lightweight 2.5D elevation map representation. However, this efficiency advantage comes at the cost of incomplete environmental characterization: the 2.5D representation inherently lacks the volumetric awareness provided by 3D methods such as TSDF and Gallant, which explains its degraded performance on terrains requiring vertical and lateral clearance reasoning (see Table III). In contrast, ADAPT achieves comparable efficiency to PIM while maintaining rich spatial perception through the dual-map design, striking a practical balance between computational overhead and perceptual completeness.

4) Ablation Study on Adaptive Mechanism: To specifically validate the obstacle avoidance capability of the adaptive mechanism, the ablation study focuses on Beam, Pole and Narrow Gate terrains where perception is critical. Besides the proposed method, two additional settings isolating the effect of radius representation and reward formulation are considered: 1) *Fixed Radius*, where the sensing radius r_t is excluded from the observation and action spaces. The policy operates with a constant radius. Separate policies are trained with $R \in \{1, 3, 5\}$ m. 2) *w/o Adaptive Reward*, where the sensing radius r_t serves as both an observation, enabling closed-loop regulation, and an action, allowing for the adjustment of perception settings. However, this configuration excludes the reward function related to r_t.

Table V summarizes the ablation results. Unlike the main benchmark which evaluates traversal through a corridor of five progressively harder obstacles, the ablation study tests each configuration against individual obstacle instances at fixed speed commands. In this single-obstacle evaluation paradigm, the normalized traversal distance (R_{trav}) is not applicable; hence only R_{succ} is reported.

A short fixed radius ($R = 1$) yields the lowest average success rates, particularly at higher speeds, indicating that the limited look-ahead range is insufficient for these terrains. A long fixed radius ($R = 5$) performs better than $R = 1$ at high speeds but remains inferior to a moderate fixed radius ($R = 3$), suggesting a trade-off between larger look-ahead range and reduced local resolution. However, the performance of a fixed radius is still behind the fully adaptive method. On the other hand, if the adaptive reward is removed as in w/o Adaptive Reward, the performance can drastically drop as speed increases, indicating that treating the sensing radius as an action alone is not sufficient to recover the best locomotion behavior. Overall, the fully adaptive method achieves the highest average success rate, indicating efficacy of the proposed closed-loop radius modulation.

C. Real-world Experiments

1) Deployment: The trained policy by ADAPT is directly deployed onto the physical Unitree G1 robot without further fine-tuning. As illustrated in Fig. 4, the policy operates across all seven *BarrierTrack* terrains in the real world. In contrast to the simulation evaluation, the real-world trials use a fixed obstacle configuration for each terrain type without progressive difficulty escalation. Each terrain is tested under

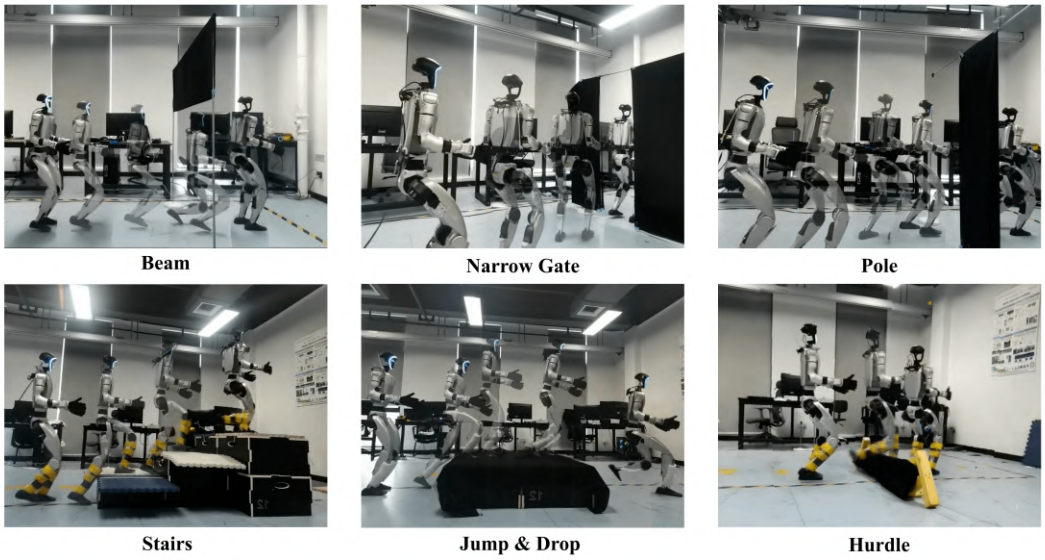


Fig. 4: Zero-shot real-world deployment across diverse terrains. The trained policy enables the physical robot to successfully traverse seven distinct *BarrierTrack* terrains without any real-world fine-tuning.

TABLE V: Ablation on adaptive sensing radius. R_{succ}: success rate (%). Values are averaged over Beam, Pole, and Narrow Gate terrains.

Type	Config	v=0.5 m/s	v=1.0 m/s	v=1.5 m/s
Fixed Radius	$R = 1$	51.33 (±17.65)	51.33 (±23.49)	24.67 (±22.32)
	$R = 3$	81.33 (±21.25)	72.67 (±15.90)	56.67 (±29.70)
	$R = 5$	64.00 (±18.90)	64.67 (±26.17)	53.33 (±30.26)
w/o Adap. Reward	no r_{adap}	92.67 (±10.93)	71.33 (±25.66)	42.67 (±29.09)
Adaptive Radius	**Ours**	**94.67** (±10.24)	**96.00** (±8.00)	**84.00** (±19.60)

three speed commands ($v \in \{0.5, 1.0, 1.5\}$ m/s), with a total of 30 trials conducted per terrain.

In terrains that require traversable-space selection, the robot identifies viable regions using the vertical distance map $\mathbf{D}_t$. When encountering aerial obstacles such as beams, the robot lowers its body to crouch underneath the beams and resumes to the normal posture afterward. For narrow pillar-like obstacles, the robot autonomously adjusts its heading to bypass the poles. When facing narrow gates, it precisely aligns its orientation with the traversable path. In terrains requiring precise foot placement, the robot relies on the elevation map $\mathbf{E}_t$ to stably ascend stairs, negotiate drops and jumps, and step over hurdles. Furthermore, because the sensing radius of ADAPT can be dynamically adjusted, the policy demonstrates robust performance even at high locomotion speeds. As shown in Table V, when navigating at a high speed of $v = 1.5$ m/s, ADAPT achieves an 84.00% success rate, significantly outperforming the best fixed-radius baseline (56.67% for $R = 3$ m). This mechanism resolves the limitations of conventional approaches constrained by fixed perception ranges against high-speed obstacle avoidance tasks, yielding superior local obstacle avoidance.

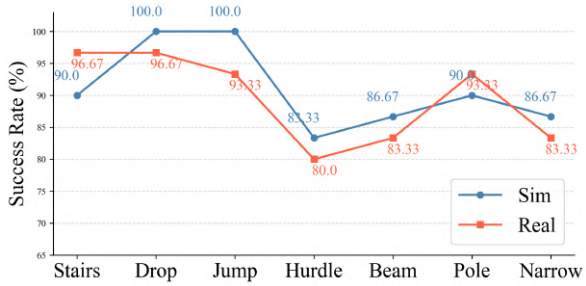


Fig. 5: Success rates across seven terrains in simulation and real-world deployment. The real-world results for each terrain are averaged over 30 trials.

2) Results: Fig. 5 presents per-terrain success rates from both simulation and physical deployment. Despite an inherent sim-to-real domain gap, the real-world success rates closely follow the trend of their counterparts in simulation, maintaining over 80% success rates across all terrains. The physical robot exhibits a range of whole-body behaviors, including stair ascent and descent, stepping across drops and gaps, crouching under beams of varying height, and navigating through narrow passages and pole fields, suggesting good generality and practical applicability of the learned policy.

V. Conclusions

This paper presented ADAPT, an adaptive dual-projection framework for perceptive humanoid traversal in complex 3D environments. By decomposing local geometry into a horizontal elevation map and a vertical distance map, ADAPT explicitly separates ground-support and body-level clearance constraints into a compact representation for high-frequency

control. Furthermore, embedding the sensing radius into the action space enables the robot to dynamically adjust its perceptual horizon based on motion context. Experiments demonstrate that ADAPT significantly improves traversal robustness and computational efficiency over fixed-range and voxel-based alternatives, achieving successful zero-shot sim-to-real transfer on a Unitree G1 robot.

Despite these capabilities, the current framework focuses solely on geometric traversability and adapts only the sensing radius. Future work will investigate richer active perception policies that jointly optimize sensing scale, resolution, and update frequency, while incorporating semantic reasoning to enable long-horizon navigation in complex, human-centered environments.

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